Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the boxes is helpful for understanding the concept(s).

21.1 Business Cycle

- A **business cycle** refers to the short run fluctuations of economic activity about its long run trend.
- There are four phases of a typical business cycle:

	Trough (depression)	Recovery (expansion)	Peak (boom/ prosperity)	Recession (contraction)
Business confidence, levels of investment, consumption and production	low	rising	high	falling
Unemployment rate	high	falling	low	rising
Real GDP	relatively low	rising	relatively high	falling
General price level	relatively low	rising	relatively high	rising moderately or even falling

21.2 Inflation and Deflation

Meanings of inflation and deflation

- The **general price level** is the average level of prices of goods and services in an economy.
  - Changes in the general price level affect people's standard of living.
- **Inflation** is a **persistent increase in the general price level**.
- **Deflation** is a **persistent decrease in the general price level**.

A temporary increase in the price of a single good does not imply the occurrence of inflation.

Disinflation refers to a decline in inflation rate, it is different from deflation.

Measurements of the general price level

- **Consumer Price Index (CPI)**
  - It is a measure of the weighted average of prices of a basket of consumer goods and services generally purchased by households to reflect changes in the general price level.
  - Increases (decreases) in the CPIs reflect an increase (a decrease) in the general price level.
  - The CPIs are set to be 100 for the base year.

When the CPI increases, the prices of some goods covered under CPI may remain unchanged or even decrease.

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- **Implicit price deflator of GDP**
  - It is a measure of the change in prices of final goods and services included in the calculation of GDP.

$$\text{Implicit price deflator of GDP} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

  - It has a value of 100 in the base year.
- Comparison between the Consumer Price Index and the implicit price deflator of GDP

  - The implicit price deflator of GDP is preferred as a measure of changes in the general price level with its more comprehensive coverage. It **includes both consumer goods and producer goods**.
  - The CPI is more often used to analyse the change in the cost of living as it only reflects changes in prices of consumer goods and services.
  - The **baskets and weights of goods and services calculated by the CPI are fixed** while **variable baskets of goods are calculated and variable weights are used in calculating implicit price deflator of GDP**.



The CPIs only include consumer goods.

- Calculation of percentage change in the general price level

$$\text{Percentage change in the general price level} = \frac{\text{Price index in the current period} - \text{Price index in the previous period}}{\text{Price index in the previous period}} \times 100\%$$

  - If the percentage change in the general price level is positive (negative), it means that there is inflation (deflation).

Effects of inflation and deflation

- During inflation (deflation):
  - the purchasing power of money decreases (increases).
  - the cost of living increases (decreases).
  - people hold fewer (more) monetary assets and more (fewer) real assets.
  - the quantity of exports decreases (increases) and the quantity of imports increases (decreases).
  - government tax revenue increases (decreases).

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- Redistributive effects of **unanticipated inflation** and **unanticipated deflation** on wealth

Anticipated inflation	Unanticipated inflation
It refers to the inflation in which people can correctly foresee the rate of the increase in the general price level.	It refers to the inflation in which people cannot correctly foresee the rate of the increase in the general price level.

  - During unanticipated inflation (deflation):
    - People who have monetary assets lose (gain) while those who have monetary liabilities gain (lose).
    - Payers of a given amount of future payments gain (lose) while payees lose (gain).

Relationship between nominal and real interest rates

- The Fisher equation

$$\text{Nominal interest rate} = \text{Real interest rate} + \text{Expected inflation rate}$$

Deflation can be interpreted as negative inflation, therefore:
Nominal interest rate = Real interest rate – Expected deflation rate.

- Illustrating wealth redistribution by comparing expected and actual (realised) real interest rates

$$\text{Expected real interest rate} = \text{Nominal interest rate} - \text{Expected inflation rate}$$

$$\text{Actual real interest rate} = \text{Nominal interest rate} - \text{Actual inflation rate}$$

			Redistributive effect to lenders	Redistributive effect to borrowers
Expected inflation rate < Actual inflation rate	→	Expected real interest rate > Actual real interest rate	lose	gain
Expected inflation rate > Actual inflation rate	→	Expected real interest rate < Actual real interest rate	gain	lose
Expected inflation rate = Actual inflation rate	→	Expected real interest rate = Actual real interest rate	no effect	no effect

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- Cost of holding money and nominal and real rates of return on holding money
  - Nominal interest rate = (Opportunity) cost of holding money
  - (Opportunity) cost of holding money = Real interest rate + Expected inflation rate
  - Nominal rate of return on holding money = 0%
  - Real rate of return on holding money
    = Nominal rate of return on holding money – Actual inflation rate

21.3 Inflation and the Quantity Theory of Money

- Equation of exchange
  - The equation:

$$MV \equiv PY$$

where:
M is **the money stock (i.e. money supply) in an economy**.
V is **the velocity of circulation of money**.
P is **the general price level**.
Y is **the total output (i.e. national output or real GDP) of an economy**.

$$\%\ \text{change in M} + \%\ \text{change in V} \approx \%\ \text{change in P} + \%\ \text{change in Y}$$

where:
'% change in M' is the growth rate of money stock (or the growth rate of money supply).
'% change in V' is the percentage change in the velocity of circulation of money.
'% change in P' is the percentage change in the general price level.
'% change in Y' is the percentage change in the total output.

  - **Velocity of circulation of money**: the average number of times a unit of money is used in a period of time to buy goods and services that make up GDP.



The velocity of circulation of money is affected by various factors including:
- consumption patterns
- technology of transactions
- income payment frequency

$$V = \frac{PY}{M}$$

  - Features of the equation of exchange:
    - The equation of exchange is an identity
    - No cause-and-effect relationship between the variables

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- Assumptions of the quantity theory of money
  - The velocity of circulation of money is a constant
  - Total output may vary in the short run but remains constant in the long run

In the short run, V is constant. In the long run, both V and Y are constant.

- Short run adjustment to changes in money supply as predicted by the quantity theory of money

$$M\bar{V} = PY$$

$$\%\text{ change in } M \approx \%\text{ change in nominal GDP } (PY)$$
$$\approx \%\text{ change in } P + \%\text{ change in } Y$$

  - Possible outcomes:

Situation	Result
% change in M > % change in Y	• % change in P > 0 • There will be an increase in the general price level.
% change in M < % change in Y	• % change in P < 0 • There will be a decrease in the general price level.
% change in M = % change in Y	• % change in P = 0 • There will be no change in the general price level.

  - An increase (a decrease) in money supply will lead to an increase (a decrease) in the general price level or real output, or increases (decreases) in both.
- Long run adjustment to changes in money supply as predicted by the quantity theory of money (Classical QTM)

$$M\bar{V} = P\bar{Y}$$

$$\%\text{ change in } M \approx \%\text{ change in } P$$

- Predictions of the quantity theory of money – 'Money is neutral'
  - A change in money supply will not affect the level of real output in the long run.

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21.4 Unemployment and Underemployment

The labour force, employed, unemployed and underemployed

- **Labour force**: is made up of the employed and the unemployed.

$$\text{Labour force} = \begin{matrix}\text{Employed population} \\ \text{(underemployed population included)}\end{matrix} + \begin{matrix}\text{Unemployed} \\ \text{population}\end{matrix}$$

- **The employed**: those who have been at work for pay or profit.
- **The unemployed**: those who want a job but fail to find one.
- **The underemployed**: those who involuntarily work less than specified working hours.

Unemployment rate and underemployment rate

$$\text{Unemployment rate} = \frac{\text{Unemployed population}}{\text{Labour force}} \times 100\%$$

$$\text{Underemployment rate} = \frac{\text{Underemployed population}}{\text{Labour force}} \times 100\%$$

The percentage change in unemployed population relative to the percentage change in labour force determines the change in the unemployment rate.

Costs of unemployment

- To the unemployed
  - lower standard of living
  - loss of skills
  - psychological problems
- To the government
  - financial burden increases
  - less salaries tax collected
- To society
  - loss in output
  - social and political unrest
  - loss of human capital

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